HAT MATRIX

Q1) What are the five properties of $h_{ii} = e_i^T H e_i$?

Q2) Prove Theorem 2 (i.e., five properties discussed in the previous question)

(a) $\frac{1}{n} \le h_{ii} \le 1$

(b) $-\frac{1}{2} \le h_{ij} \le \frac{1}{2}$

(c) $\sum_{i=1}^{n} h_{ii} = p+1$

(d) $\sum_{i=1}^{n} h_{ij} = 1$

(e) $h_{ii} = \frac{1}{n} + (z_i - \bar{x})^T (X_{1,\perp}^T X_{1,\perp})^{-1} (z_i - \bar{x})$, $z_i = (x_{i1} \cdots x_{ip})^T$, $\bar{x} = (\bar{x}_1 \cdots \bar{x}_p)^T$

(Solution)

(e)

$$\begin{aligned} h_{ii} &= e_i^T \Pi_X e_i = e_i^T (\Pi_{\mathbb{1}} + \Pi_{X_{1,\perp}}) e_i \\ &= e_i^T \Pi_{\mathbb{1}} e_i + e_i^T \Pi_{X_{1,\perp}} e_i \\ &= e_i^T \mathbb{1} (\mathbb{1}^T \mathbb{1})^{-1} \mathbb{1}^T e_i + e_i^T X_{1,\perp} (X_{1,\perp}^T X_{1,\perp})^{-1} X_{1,\perp}^T e_i \\ &= \frac{1}{n} \cdot 1 \cdot 1 + (z_i - \bar{x})^T (X_{1,\perp}^T X_{1,\perp})^{-1} (z_i - \bar{x}) \\ &= \frac{1}{n} + (z_i - \bar{x})^T (X_{1,\perp}^T X_{1,\perp})^{-1} (z_i - \bar{x}) \end{aligned}$$

(a) $(X_{1,\perp}^T X_{1,\perp})^{-1}$ is positive definite because $X_{1,\perp}^T X_{1,\perp}$ is positive definite.

$\therefore (z_i - \bar{x})^T (X_{1,\perp}^T X_{1,\perp})^{-1} (z_i - \bar{x}) \ge 0$

$\Rightarrow h_{ii} \ge 1/n + 0$

$$\begin{aligned} e_i^T H e_i &= e_i^T H^T H e_i = (He_i)^T (He_i) = (e_i^T H)(e_i^T H)^T \\ &= (h_{ij})_{1 \le j \le n} (h_{ji})_{1 \le j \le 1} \\ &= \sum_{j=1}^{n} h_{ij} \cdot h_{ji} = \sum_{j=1}^{n} h_{ij}^2 = h_{ii} \end{aligned}$$

$\Rightarrow h_{ii}^2 \le h_{ii}$

$\therefore h_{ii} \le 1$

$\therefore \frac{1}{n} \le h_{ii} \le 1$

(b) $-\frac{1}{2} \le h_{ij} \le \frac{1}{2}$

$h_{ii} = e_i^T H e_i = \sum_{k=1}^{n} h_{ik}^2$, $\quad \therefore h_{ii} \ge h_{ii}^2 + h_{ij}^2$

$\Rightarrow h_{ii}(1 - h_{ii})^2 \ge h_{ij}^2$

$\Rightarrow \frac{1}{4} \ge h_{ij}^2$

$\therefore -\frac{1}{2} \le h_{ij} \le \frac{1}{2}$

(c) $\sum_{i=1}^{n} h_{ii} = p+1$

$$\begin{aligned} \sum_{i=1}^{n} h_{ii} &= \text{trace}(H) = \text{trace}(X(X^TX)^{-1}X^T) = \text{trace}(X^TX(X^TX)^{-1}) \\ &= \text{trace}(I_{p+1}) = p+1 \end{aligned}$$

$\therefore \sum_{i=1}^{n} h_{ii} = p+1$

(d)

$$\begin{aligned} \sum_{i=1}^{n} h_{ij} &= [1 \cdots 1] \begin{bmatrix} h_{1j} \\ \vdots \\ h_{nj} \end{bmatrix} = \mathbb{1}^T H e_i = \mathbb{1}^T (\Pi_{\mathbb{1}} + \Pi_{X_{1,\perp}}) e_i \\ &= \mathbb{1}^T \Pi_{\mathbb{1}} e_i \\ &= \mathbb{1}^T \mathbb{1} (\mathbb{1}^T \mathbb{1})^{-1} \mathbb{1}^T e_i \\ &= \mathbb{1}^T e_i = 1 \end{aligned}$$

Leverage and Influential Observation

Q3) Define Leverage:

h_{ii} is called the "leverage" of the i^{th} observation

Q4) What is the implication of large leverage? i.e., h_{ii} ?
Mathematically show that large leverage implies the following:
$\hat{y}_i \simeq y_i$

$$h_{ii} = \frac{1}{n} + (z_i - \bar{x})^T (X_{1,\perp}^T X_{1,\perp})^{-1} (z_i - \bar{x})$$

$\therefore$ large h_{ii} implies that z_i (i^{th} observation) is far from $\bar{x}$

In the simple linear regression case:

$y_i = \beta_0 + \beta_1 x_i + \varepsilon_i$, $\varepsilon_i \sim N(0, \sigma^2)$

$X = (\mathbb{1} \ x)$, $x = [x_1 \cdots x_n]^T$

$$h_{ii} = \frac{1}{n} + \frac{(x_i - \bar{x})^2}{S_{xx}}$$

For any given $i = 1, \cdots, n$ we have already shown that:

$\hat{y} = Hy$ and hence $\hat{y}_i = h_{ii} y_i + \sum_{i=1}^{n} h_{ij} y_j$

if $h_{ii} \simeq 1$, y_i contributes more to $\hat{y}_i$ than y_j ($j = 1, \cdots, n, j \ne i$)

Meaning, $\hat{y}_i \approx y_i$

This implies that the estimated regression function will be close to y_i; that is, $\hat{y}_i - y_i$ will be small.

▷ Notice that large h_{ii} indicates that the i^{th} data point has a smaller variance for its residual since

$$\begin{aligned} Var(\hat{\varepsilon}_i) = Var(y_i - \hat{y}_i) &= Var(e_i^T(y - Hy)e_i) \\ &= e_i^T (I - H) Var(y) e_i \\ &= \sigma^2 (e_i^T (I - H) e_i) \\ &= \sigma^2 (1 - h_{ii}) \end{aligned}$$

Q5) What is an "Influential" point?

An implication point refers to the i^{th} observation such that two regression fits, one with and one without it, gives quite different results.

Q6) What is a common criterion that is used to identify a leverage point?

If the leverage of the i^{th} observation, h_{ii}, is greater than $2(p+1)/n$, meaning:

$$h_{ii} > \frac{2(p+1)}{n},$$

i^{th} observation is called a leverage point.

Q7) Are leverage points influential points?

Not necessarily.

Leverage points are solely determined by the design matrix X and influential points are identified after fitting the regression model using the design matrix with and without the point of interest.

Q8) What is the difference between a leverage and an influential point?

Answered above

COOK DISTANCE

Q9) Define the Cook Distance.

$$D_i = \frac{(\hat{y}_{(i)} - \hat{y})^T(\hat{y}_{(i)} - \hat{y})}{(p+1)\hat{\sigma}^2} = \frac{(\hat{\beta}_{(i)} - \hat{\beta})^T X^T X(\hat{\beta}_{(i)} - \hat{\beta})}{(p+1)\hat{\sigma}^2}$$

* the subscript (i) in $\hat{y}_{(i)}$ and $\hat{\beta}_{(i)}$ denotes the estimator with the i^{th} observation deleted.

Q10) Prove Lemma 6

Let $z_i = (1, x_{i1} \cdots x_{ip})^T$ denote the i^{th} observation (observed predictor).

$$\hat{\beta}_{(i)} = \hat{\beta} - \frac{\hat{\varepsilon}_i}{1-h_{ii}}(X^TX)^{-1}z_i \quad (i = 1, \cdots, n)$$

$$\hat{\beta}_{(i)} = (X_{(i)}^T X_{(i)})^{-1} X_{(i)}^T y_{(i)}, \quad X = [z_1 \cdots z_n]^T$$

$$X_{(i)}^T X_{(i)} = X^TX - z_i z_i^T, \quad (X^TX - z_i z_i^T)^{-1} = (X^TX)^{-1} + \frac{(X^TX)^{-1} z_i z_i^T (X^TX)^{-1}}{1 - z_i^T (X^TX)^{-1} z_i}$$

$$X_{(i)}^T y_{(i)} = X^T y - z_i y_i$$

$$\therefore (X_{(i)}^T X_{(i)})^{-1} X_{(i)}^T y_{(i)} = \left[(X^TX)^{-1} + \frac{(X^TX)^{-1} z_i z_i^T (X^TX)^{-1}}{1 - z_i^T (X^TX)^{-1} z_i}\right]\left[X^T y - z_i y_i\right]$$

$$= \hat{\beta} - (X^TX)^{-1} z_i y_i + \frac{(X^TX)^{-1} z_i z_i^T \hat{\beta}}{1 - z_i^T (X^TX)^{-1} z_i} - \frac{(X^TX)^{-1} z_i z_i^T (X^TX)^{-1} z_i y_i}{1 - z_i^T (X^TX)^{-1} z_i}$$

$$= \hat{\beta} - (X^TX)^{-1} z_i y_i + \frac{(X^TX)^{-1} z_i \hat{y}_i}{1 - h_{ii}} - \frac{(X^TX)^{-1} z_i h_{ii} y_i}{1 - h_{ii}}$$

$$= \hat{\beta} - \frac{(X^TX)^{-1} z_i}{1 - h_{ii}}\left\{(1 - h_{ii}) y_i - \hat{y}_i + h_{ii} y_i\right\}$$

$$= \hat{\beta} - \frac{(X^TX)^{-1} z_i}{1 - h_{ii}}\left\{y_i - \hat{y}_i\right\}$$

$$= \hat{\beta} - \frac{(X^TX)^{-1} z_i \hat{\varepsilon}_i}{1 - h_{ii}}$$

Q11) What is another respresentation of the cook Distance?

$$D_i = \frac{(\hat{\beta} - \hat{\beta}_{(i)})^T X^T X (\hat{\beta} - \hat{\beta}_{(i)})}{(p+1)\hat{\sigma}^2} = \frac{\{(X^TX)^{-1} z_i \hat{\varepsilon}_i\}^T X^T X \{(X^TX)^{-1} z_i \hat{\varepsilon}_i\}}{(1 - h_{ii})^2 (p+1)\hat{\sigma}^2}$$

$$= \frac{\hat{\varepsilon}_i^2 z_i^T (X^TX)^{-1} z_i}{(1 + h_{ii})^2 (p+1)\hat{\sigma}^2} = \frac{\varepsilon_i^2 h_{ii}}{(1 + h_{ii})^2 (p+1)\hat{\sigma}^2} = \frac{\varepsilon_i^2}{(p+1)\hat{\sigma}^2} \cdot \frac{h_{ii}}{(1 + h_{ii})^2}$$

Q12) Prove Theorem 7

$$D_i = \frac{\varepsilon_i^2}{(p+1)\hat{\sigma}^2} \cdot \frac{h_{ii}}{(1 + h_{ii})^2} \quad (i = 1, 2, \cdots, n)$$

RESIDUALS

Q13) Define the (internally) studentized residual $\hat{\gamma}_i$ and the (externally) studentized residual $\hat{t}_i$.

$$\hat{\gamma}_i = \frac{\hat{\varepsilon}_i}{\hat{\sigma}\sqrt{1-h_{ii}}}, \quad \hat{t}_i = \frac{\hat{\varepsilon}_i}{\hat{\sigma}_{(i)}^2\sqrt{1-h_{ii}}} \quad (i=1,\cdots,n)$$

$* \ Var(\hat{\varepsilon}) = Var((I_n-\Pi_X)y) = (I_n-\Pi_X)Var(y) = (I_n-\Pi_X)\sigma^2$

$Var(\hat{\varepsilon}_i) = Var(e_i^T\hat{\varepsilon}) = e_i^T(I_n-\Pi_X)e_i\cdot\sigma^2 = \sigma^2(1-h_{ii})$

* The subscript (i) denotes the estimator with the i^{th} observation being deleted.

More specifically, we define $z_j = (1, x_{j1} \cdots x_{jp})^T$ so that

$$\sigma_{(i)}^2 = \frac{1}{n-p-2}\sum_{j\neq i}(y_j-\hat{y}_{j,(i)})^2 \quad \text{with} \quad \hat{y}_{j,(i)} = z_j^T\hat{\beta}_{(i)}$$

$$\sigma^2 = \frac{1}{n-p-1}\sum_{j=1}^n(y_j-\hat{y}_j)^2 \quad \text{with} \quad \hat{y}_j = z_j^T\hat{\beta}$$

Q14) Prove Lemma 10

$$y_j - \hat{y}_{j,(i)} = \hat{\varepsilon}_j + \frac{h_{ji}\hat{\varepsilon}_i}{1-h_{ii}}$$

$$\begin{aligned} y_j - \hat{y}_{j,(i)} = y_j - z_j^T\hat{\beta}_{(i)} &= y_j - z_j^T\left(\hat{\beta} - \frac{\hat{\varepsilon}_i(X^TX)^{-1}z_i}{1-h_{ii}}\right) \\ &= y_j - z_j^T\hat{\beta} + \frac{\hat{\varepsilon}_i z_j^T(X^TX)^{-1}z_i}{1-h_{ii}} \\ &= \hat{\varepsilon}_j + \frac{\hat{\varepsilon}_i h_{ji}}{1-h_{ii}} \end{aligned}$$

Q15) Prove the following:

$$(n-p-2)\hat{\sigma}_{(i)}^2 = (n-p-1)\hat{\sigma}^2 + \frac{\hat{\varepsilon}_i^2}{1-h_{ii}}$$

$$\begin{aligned} (n-p-2)\hat{\sigma}_{(i)}^2 &= \sum_{j\neq i}(y_j-\hat{y}_{j,(i)})^2 = \sum_{j=1}^n(y_j-\hat{y}_{j,(i)})^2 - (y_i-\hat{y}_{i,(i)})^2 \\ &= \sum_{j=1}^n\left(\hat{\varepsilon}_j - \frac{\hat{\varepsilon}_i h_{ji}}{1-h_{ii}}\right)^2 - \left(\hat{\varepsilon}_i + \frac{\hat{\varepsilon}_i h_{ii}}{1-h_{ii}}\right)^2 \\ &= \sum_{j=1}^n\left\{\hat{\varepsilon}_j^2 - 2\cdot\frac{\hat{\varepsilon}_j\hat{\varepsilon}_i h_{ji}}{1-h_{ii}} + \left(\frac{\hat{\varepsilon}_i h_{ji}}{1-h_{ii}}\right)^2\right\} - \left(\frac{\hat{\varepsilon}_i - \hat{\varepsilon}_i h_{ii} + \hat{\varepsilon}_i h_{ii}}{1-h_{ii}}\right)^2 \\ &= \sum_{j=1}^n\hat{\varepsilon}_j^2 + \frac{\hat{\varepsilon}_i^2}{(1-h_{ii})^2}\sum_{j=1}^n h_{ji}^2 - \frac{\varepsilon_i^2}{(1-h_{ii})^2} \\ &= \sum_{j=1}^n\hat{\varepsilon}_j^2 - \frac{\varepsilon_i^2(1-h_{ii})}{(1-h_{ii})^2} = \sum_{j=1}^n\varepsilon_j^2 - \frac{\varepsilon_i^2}{1-h_{ii}} \quad \left(\because \sum_{i=1}^n h_{ji}^2 = e_i^TH^2e_i = e_i^THe_i = h_{ii}\right) \end{aligned}$$

Q16) Prove Theorem 11

$$\frac{\hat{\gamma}_i^2}{n-p-1} \sim Beta\left(\frac{1}{2}, \frac{n-p-2}{2}\right)$$

$$\hat{t}_i \sim t(n-p-2)$$

$\hat{\gamma}_i^2$: internally studentized residual $= \hat{\varepsilon}_i/(\hat{\sigma}\sqrt{1-h_{ii}})$

$\hat{\varepsilon}_i \sim N(0, (1-h_{ii})\sigma^2)$

$$\therefore \frac{\hat{\varepsilon}_i}{\sigma\sqrt{1-h_{ii}}} \sim N(0,1) \Rightarrow \left(\frac{\hat{\varepsilon}_i}{\sigma\sqrt{1-h_{ii}}}\right)^2 \sim \chi^2(1)$$

$$(n-p-1)\hat{\sigma}^2 = (n-p-2)\hat{\sigma}_{(i)}^2 + \frac{\hat{\varepsilon}_i^2}{1-h_{ii}}$$

$$\Rightarrow \frac{(n-p-1)\hat{\sigma}^2}{\sigma^2} = \frac{(n-p-2)\hat{\sigma}_{(i)}^2}{\sigma^2} + \left(\frac{\hat{\varepsilon}_i}{\sigma\sqrt{1-h_{ii}}}\right)^2$$

$\hat{\sigma}_{(i)}^2$ and $\hat{\varepsilon}_i$ are independent

$\Leftrightarrow \hat{\sigma}_{(i)}^2, \ y_i - z_i^T\hat{\beta}_{(i)}$ are independent

$$\frac{(n-p-1)\hat{\sigma}^2}{\sigma^2} \sim \chi^2(n-p-1), \quad \left(\frac{\varepsilon_i}{\sigma\sqrt{1-h_{ii}}}\right)^2 \sim \chi^2(1)$$

$$\therefore \frac{(n-p-2)\hat{\sigma}_{(i)}^2}{\sigma^2} \sim \chi^2(n-p-2)$$

$$\frac{\hat{\gamma}_i^2}{n-p-1} = \frac{\dfrac{\varepsilon_i^2}{(1-h_{ii})}}{(n-p-1)\hat{\sigma}^2} = \frac{\varepsilon_i^2/\sigma^2(1-h_{ii})}{(n-p-1)\hat{\sigma}^2/\sigma^2} = \frac{V_1}{V_1+V_2}$$

$V_1 \sim \chi^2(1) \equiv Gamma(\frac{1}{2}, 2)$, $V_2 \sim \chi^2(n-p-2) \overset{d}{\equiv} Gamma(\frac{n-p-2}{2}, 2)$
& $V_1 \perp\!\!\!\perp V_2$

$$\therefore \frac{\hat{\gamma}_i^2}{n-p-1} \sim Beta\left(\frac{1}{2}, \frac{n-p-2}{2}\right)$$

$$\hat{t}_i^2 = \frac{\hat{\varepsilon}_i^2}{\sigma_{(i)}\sqrt{1-h_{ii}}} = \frac{\hat{\varepsilon}_i^2}{\sigma\sqrt{1-h_{ii}}} \bigg/ \sqrt{\frac{(n-p-2)\hat{\sigma}_{(i)}^2}{\sigma^2}\bigg/(n-p-2)}$$

$$= Z\Big/\sqrt{V_1/n_1}, \quad Z\sim N(0,1),\ V_1 \sim \chi^2(n_1),\ n_1 = n-p-1$$

$Z \perp\!\!\!\perp V_1$

$$\therefore \hat{t}_i^2 \sim t(n-p-2)$$

Q16-1) Define PRESS (Prediction Sum of Square) and show that PRESS $\geq$ SSE

$$PRESS = \sum_{i=1}^n(y_i - y_{i,(i)})^2 = \sum_{i=1}^n\left(\frac{\hat{\varepsilon}_i}{1-h_{ii}}\right)^2 \geq \sum_{i=1}^n\hat{\varepsilon}_i^2 = SSE$$

$\therefore$ PRESS $\geq$ SSE

GENERALIZED LEAST SQUARE ESTIMATION

Consider the MLRM where the errors do not have equal variance or they are not independent.

$$y = X\beta + \varepsilon \quad (E[\varepsilon] = 0,\ Var(\varepsilon) = \sigma^2 V)$$, V is symmetric and positive definite

Q17) Prove Theorem 13

$$V^{-1/2}y = V^{-1/2}X\beta + V^{-1/2}\varepsilon \quad , \ \varepsilon^* = V^{-1/2}\varepsilon,\ E[\varepsilon^*] = 0,\ Var(\varepsilon^*) = \sigma^2 I_n$$

$$E[\varepsilon^*] = E[V^{-1/2}\varepsilon] = V^{-1/2}E[\varepsilon] = 0$$

$$Var(\varepsilon^*) = Var(V^{-1/2}\varepsilon) = V^{-1/2}Var(\varepsilon)V^{-1/2} = V^{-1/2}VV^{-1/2} = V^{-1/2}V^{1/2}V^{1/2}V^{-1/2} = I_n$$

Q18) Show that:

$$\hat{\beta}_G = (X^TV^{-1}X)^{-1}X^TV^{-1}y \quad , \ E[\hat{\beta}_G] = \beta,\ Var(\hat{\beta}_G) = \sigma^2(X^TV^{-1}X)^{-1}$$

$$y^* = X^*\beta + \varepsilon^* \quad , \ y^* = V^{-1/2},\ X^* = V^{-1/2}X,\ \varepsilon^* = V^{-1/2}\varepsilon^*$$

$$\therefore \hat{\beta}_G = (X^{*T}X^*)^{-1}X^{*T}y^* = (X^TV^{-1/2}\cdot V^{-1/2}X)^{-1}X^TV^{-1/2}V^{-1/2}y$$
$$= (X^TV^{-1}X)^{-1}X^TV^{-1}y$$

$$E[\hat{\beta}_G] = (X^{*T}X^*)^{-1}X^{*T}X^*\beta = \beta$$
$$Var(\hat{\beta}_G) = \sigma^2(X^{*T}X^*)^{-1} = \sigma^2(X^TV^{-1}X)^{-1}$$

WEIGHTED LEAST SQUARE ESTIMATOR

Consider the generalized MLRM,

$y = X\beta + \varepsilon$, where $E[\varepsilon] = 0$ and $Var(\varepsilon) = \sigma^2 W$

For some diagonal matrix $W = diag(w_1^{-1}, w_2^{-2}, \cdots, w_n^{-1})$

The error terms are uncorrelated but have unequal variances

$$Var(\varepsilon_i) = \frac{\sigma^2}{w_i}$$

In this special case, the generalized least square estimator simply minimizes the weighted sum of squared errors.

$$\sum_{i=1}^{n} w_i(y_i - \beta_0 - \beta_1 \cdots - \beta_p x_{ip})^2 = (y - X\beta)^T W^{-1}(y - X\beta)$$

where the weight of each data observation is inversely proportional to the variance of the corresponding response.

Q19) Find $\hat{\beta}_W$

$$W^{-1/2}y = W^{-1/2}X\beta + W^{-1/2}\varepsilon$$
$$y^* = X^*\beta + W^{-1/2}\varepsilon$$

$$\therefore \hat{\beta}_W = (X^{*T}X^*)^{-1}X^{*T}y^* = (X^TW^{-1}X)^{-1}X^TW^{-1}y$$

$$\hat{\beta}_W = (X^TW^{-1}X)^{-1}X^TW^{-1}y = \underset{\beta}{\operatorname{argmin}} \|y^* - X^*\beta\|^2$$
$$= \|W^{-1/2}(y - X\beta)\|^2$$
$$= (y - X\beta)W^{-1}(y - X\beta)$$

BEST LINEAR UNBIASED ESTIMATOR (BLUE)

Q20) Prove the Gauss-Markov Theorem

Consider the generalized MLRM: $y = X\beta + \varepsilon$, $E[\varepsilon] = 0$, $Var(\varepsilon) = \sigma^2 V$

In this case, the generalized least square estimator $\hat{\beta}_G$ is the best among all linear unbiased estimators for β.

More specifically, let $\tilde{\beta}$ be a linear and unbiased estimator for β. The $Var(\tilde{\beta}) - Var(\hat{\beta}_G)$ is non-negative definite.

(solution)

$\tilde{\beta} = By + (X^TV^{-1}X)^{-1}X^TV^{-1}y$, $B \in \mathbb{R}^{(p+1)\times n}$

$E[\tilde{\beta}] = \beta$ $\therefore$ $BX\beta + (X^TV^{-1}X)^{-1}X^TV^{-1}X\beta$

$= BX\beta + \beta = \beta$ $\therefore$ $BX\beta = 0$ $\forall \beta \in \mathbb{R}^{p+1} \Rightarrow BX = 0$

$$
\begin{aligned}
Var(\tilde{\beta}) &= Var(By + (X^TV^{-1}X)^{-1}X^TV^{-1}y) \\
&= Var(By) + Var((X^TV^{-1}X)^{-1}X^TV^{-1}y) + 2Cov(By, (X^TV^{-1}X)^{-1}X^TV^{-1}y) \\
&= BVar(y)B^T + (X^TV^{-1}X)^{-1}X^TV^{-1}Var(y)V^{-1}X(X^TV^{-1}X)^{-1} + 2BCov(y,y)V^{-1}X(X^TV^{-1}X)^{-1} \\
&= \sigma^2BVB^T + \sigma^2(X^TV^{-1}X)^{-1} + 2\sigma^2BX(X^TV^{-1}X)^{-1} \\
&= \sigma^2BVB^T + \sigma^2(X^TV^{-1}X)^{-1}
\end{aligned}
$$

WTS: $Var(\tilde{\beta}) \geq Var(\hat{\beta}_G)$

$\therefore$ $Var(\tilde{\beta}) - Var(\hat{\beta}_G) = \sigma^2BVB^T$, $\sigma^2 x^TBVB^Tx = \sigma^2\|V^{-1/2}B^Tx\|^2 \geq 0$

$\therefore$ $Var(\tilde{\beta}) - Var(\hat{\beta}_G)$ is non-negative definite.

LACK OF FIT TEST:

Q21) Prove Theorem 15

Assume that we have $r_i \in \mathbb{Z}_+$ number of repeated measurements $y_{i,1}, \cdots, y_{i,r_i}$

$y_F = X_F + \varepsilon$, $\varepsilon_F \sim N(0, \sigma^2 I)$

$\hat{\beta} = (X_F^TX_F)^{-1}X_F^Ty = (X^TWX)^{-1}X^TW\bar{y}$

$y = [\bar{y}_1 \cdots \bar{y}_n]^T$, $\bar{y}_i = \frac{1}{n}\sum_{j=1}^{r_i} y_{ij}$

$$
X = \begin{bmatrix} 1 & x_{11} & \cdots & x_{1p} \\ \vdots & & & \\ 1 & x_{n1} & \cdots & x_{np} \end{bmatrix} = \begin{bmatrix} z_1^T \\ \vdots \\ z_n^T \end{bmatrix} , \quad X_F = \begin{bmatrix} \left.\begin{matrix} z_1^T \\ \vdots \\ z_1^T \end{matrix}\right\} r_1\text{개} \\ \vdots \\ \left.\begin{matrix} z_n^T \\ \vdots \\ z_n^T \end{matrix}\right\} r_n\text{개} \end{bmatrix}
$$

$$
\begin{aligned}
\hat{\beta} = (X_F^TX_F)^{-1}X_F^Ty_F &= \underset{\beta \in \mathbb{R}^{p+1}}{\operatorname{argmin}} \sum_{i=1}^{n}\sum_{j=1}^{r_i} \|y_{ij} - z_i^T\beta\|^2 \\
&= \underset{\beta \in \mathbb{R}^{p+1}}{\operatorname{argmin}} \sum_{i=1}^{n}\sum_{j=1}^{r_i} (y_{ij} - \bar{y}_i + \bar{y}_i - z_i^T\beta)^2 \\
&= \underset{\beta \in \mathbb{R}^{p+1}}{\operatorname{argmin}} \sum_{i,j} (y_{ij} - \bar{y}_i)^2 + (y_{ij} - \bar{y}_i)(\bar{y}_i - z_i^T\beta) + (\bar{y}_i - z_i^T\beta)^2 \\
&= \underset{\beta \in \mathbb{R}^{p+1}}{\operatorname{argmin}} \sum_{i=1}^{n}\sum_{j=1}^{r_i} (\bar{y}_i - z_i^T\beta)^2 \\
&= \underset{\beta \in \mathbb{R}^{p+1}}{\operatorname{argmin}} \sum_{i=1}^{n} r_i(\bar{y}_i - z_i^T\beta)^2 \\
&= (X^TW^{-1}X)^{-1}X^TW^{-1}\bar{y}
\end{aligned}
$$

Q22) Prove Theorem 16

$$
\begin{aligned}
\sum_{i=1}^{n}\sum_{j=1}^{r_i}(y_{ij} - \hat{y}_i)^2 &= \sum_{i,j}(y_{ij} - \bar{y}_i + \bar{y}_i - \hat{y}_i)^2 \\
&= \sum_{i,j}(y_{ij} - \bar{y}_i)^2 + (y_{ij} - \bar{y}_i)(\bar{y}_i - \hat{y}_i) + (\bar{y}_i - \hat{y}_i)^2 \\
&= \sum_{i,j}(y_{ij} - \bar{y}_i)^2 + (\bar{y}_i - \hat{y}_i)^2 \quad \because \sum_{i=1}^{n}\sum_{j=1}^{r_i}(y_{ij} - \bar{y}_i)(\bar{y}_i - \hat{y}_i) = 0
\end{aligned}
$$

Q23) Prove Theorem 17

$SSLF/\sigma^2 \sim \chi^2(n-p-1)$ SSLF : Sum Squared Lack of Fit

$SSPE/\sigma^2 \sim \chi^2(N-n)$, SSPE : Sum Squared Pure Error

SSLF and SSPE are independent

$$SSPE = \sum_{i=1}^{n}\sum_{j=1}^{r_i}(y_{ij}-\bar{y}_i)^2 = y_F^T(I_n - \Pi_{X_P})y_F \;,\; X_P = \begin{bmatrix} \mathbb{1}_{r_1} & 0 & \cdots & 0 \\ 0 & \mathbb{1}_{r_2} & \cdots & 0 \\ 0 & \cdots & \cdots & \mathbb{1}_{r_n} \end{bmatrix}$$

$$SSLF = \sum_{i=1}^{n}\sum_{j=1}^{r_i}(\bar{y}_i - \hat{y}_i)^2 = y_F^T(\Pi_{X_P} - \Pi_{X_F})y_F$$

$$SSPE/\sigma^2 = y_F^T\left(\frac{I_N - \Pi_{X_P}}{\sigma^2}\right)y_F = y^TAy$$

$A\Sigma = \left(\frac{I_n - \Pi_{X_P}}{\sigma^2}\right)\sigma^2 I_N = I_N - \Pi_{X_P}$ $\therefore$ $A\Sigma$ is idempotent

non-centrality parameter, $\delta = E[y_F]^T A E[y_F] = \beta^T X_F^T (I_n - \Pi_{X_P}) X_F^T \beta = 0$

$\because \Pi_{X_P} X_F = X_F$

$r = rank(I_N - \Pi_{X_P}) = trace(I_N - \Pi_{X_P}) = N - n$, $N = \sum_{i=1}^{n} r_i$

* Note that each column vector in X_F can be produced by a linear combination of the column vectors in X_P. $\Rightarrow C(X_F) \subset C(X_P)$

$\therefore SSPE \sim \chi^2(N-n)$

$$SSLF = y_F^T(\Pi_P - \Pi_F)y_F/\sigma^2 = y^TBy \;,\; B = \left(\frac{\Pi_P - \Pi_F}{\sigma^2}\right)$$

$B\Sigma = \left(\frac{\Pi_P - \Pi_F}{\sigma^2}\right)\sigma^2 I_N = \Pi_P - \Pi_F$: idempotent

$\Rightarrow$ ① $(\Pi_P - \Pi_F) = \Pi_P^2 - \Pi_P\Pi_F - \Pi_F\Pi_P + \Pi_F^2 = \Pi_P - \Pi_F - \Pi_F + \Pi_F = \Pi_P - \Pi_F$

$\Rightarrow$ ② $(\Pi_P - \Pi_F)^T = \Pi_P^T - \Pi_F^T = \Pi_P - \Pi_F$

$\therefore \Pi_P - \Pi_F$ is a projection

non-centrality parameter δ: $E[y_F]^T B E[y_F] = \beta^T X_F^T\left(\frac{\Pi_P - \Pi_F}{\sigma^2}\right)X_F\beta = 0$

$\because (\Pi_P - \Pi_F)X_F = X_F - X_F = 0$

$r = rank(\Pi_P - \Pi_F) = n - (p+1) = n-p-1$

$\therefore SSLF/\sigma^2 \sim \chi^2(n-p-1)$

Independence between SSLF and SSPE :

$$A\Sigma B = \left(\frac{I_N - \Pi_P}{\sigma^2}\right)\cdot\sigma^2 I_N\left(\frac{\Pi_P - \Pi_F}{\sigma^2}\right) = \sigma^{-2}(I_N - \Pi_P)(\Pi_P - \Pi_F)$$
$$= \sigma^{-2}(\Pi_P - \Pi_P - \Pi_F + \Pi_F) = 0$$

$\therefore$ SSPE and SSLF are independent.

Q24) Prove Theorem 18

Consider a test statistic $F = \frac{SSLF/(n-p-1)}{SSPE/(N-n)}$. Show that, under H_0, $F \sim F(n-p-1, N-n)$

$$F = \frac{SSLF/(n-p-1)}{SSPE/(N-n)} = \frac{V_1/(n-p-1)}{V_2/(N-n)} \;,\; V_1 = \frac{SSLF}{\sigma^2} \sim \chi^2(n-p-1)$$
$$V_2 = \frac{SSPE}{\sigma^2} \sim \chi^2(N-n)$$

$\therefore F \sim F(n-p-1, N-n)$